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AFFINE PROJECTION ALGORITHM FOR ACOUSTIC FEEDBACK CANCELLATION USING PREDICTION ERROR METHOD IN HEARING AIDS

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ABSTRACT

Prediction error method (PEM) is popularly applied to acoustic feedback cancellation (AFC) in hearing aids. Commonly, this method uses normalized least mean square (NLMS) adaptive filter to estimate the coefficients of the real feedback path. A disadvantage of NLMS algorithm is to provide a slow convergence rate when coloured incoming signals are used. To address this problem, we propose a simple but effective way to apply an affine projection algorithm (APA) to acoustic feedback cancellation using PEM. Performance of the proposed method is evaluated for speech incoming signal in both cases of using/not using a probe noise. Simulation results show that the proposed method outperforms the PEM using NLMS in both terms of misalignment and added stable gain.

Index Terms— Acoustic feedback cancellation, hearing aids, NLMS, APA, PEM.

1. INTRODUCTION

Hearing aids are often prone to acoustic feedback which is formed by coupling a partly loudspeaker signal into the microphone. This phenomenon deteriorates the intelligibility of the sound as well as reduces the achievable maximum stable gain in hearing aids. Under some circumstances, "howling" may occur. Recently, acoustic feedback cancellation (AFC) is a dominant method to cancel the feedback signal. In this method, an adaptive filter is employed in the feedback canceller's path to estimate the real feedback path, then subtracts it from the microphone signal. However, the correlation between the incoming and loudspeaker signals yields a bias in the estimation of the feedback path.

In the past ten years, acoustic feedback cancellation methods employing prediction filters have been widely used. In these methods, the microphone and loudspeaker signals were pre-filtered by an inverse incoming signal model before being fed into an adaptive filter [1, 2, 3, 4]. As a result, less correlation between the loudspeaker and the incoming signal is achieved, and the bias problem in estimation of the feedback path can be reduced significantly.

In common, the well-known NLMS algorithm is found to be an attractive choice for the AFC-PEM due to its simplicity and efficiency. However, this algorithm tends to deteriorate the convergence rate of the system when the incoming signals are spectrally coloured. A potential solution to that problem is to use the affine projection algorithms [5, 6, 7] in which the current as well as the previous input vectors are used to estimate the adaptive filter's coefficients. Consequently, the APA can provide a superior convergence rate compared to NLMS algorithm, especially for speech incoming signals.

The APA and its modifications have been widely employed for AEC [7, 8, 9, 10, 11]. However, their applications for AFC in hearing aids so far has been limited. For example, in [12] a modified pseudo APA based AFC was investigated. However, this method only mentioned the application for a speech-shaped AR noise input, not for a speech incoming signal. Also, a large projection order of 20 was chosen for the APA. Note that a large order of the APA considerably increases computational complexity. In [13] an approximated APA based on Gauss-Seidel iteration for AFC was introduced. This approximated APA method was only evaluated with an added probe noise. We know that a probe noise coupled in the loudspeaker signal will lower the correlation, but a strong probe signal might disturb the hearing aid users when it becomes audible. Besides, this method does not seem to yield a good performance for the case of speech incoming signals, even a quite high affine projection order (P = 4) was used. The APA and a variable Gaussian step-size control for APA were applied to two-microphone AFC [14]. This method provides a significant improvement in performance for speech signals, but a second microphone is required.

In this paper, we propose an AFC method using a single microphone and a single loudspeaker, PEM-APA, which uses the PEM to decorrelate the input signals as well as implements a classical APA to significantly improve the convergence rate and tracking rate and to lower the steady-state error. In the proposed method, the smallest affine projection order (P = 2) is chosen to ensure only a minor increase in computational complexity compared to PEM using NLMS (PEM-NLMS). The probe noise might not be added into the loudspeaker signal. The simulation results show that the proposed method outperforms the PEM-NLMS even in the case of no probe noise. When a probe noise is injected, a further gain in both terms of misalignment and added stable gain is achieved.

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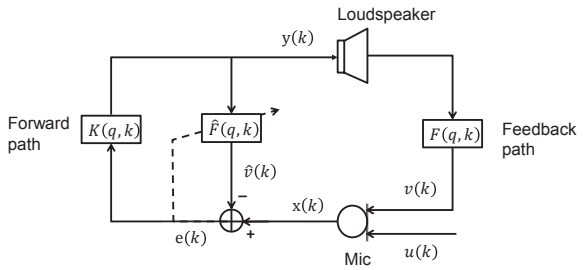


Fig. 1: A traditional AFC model

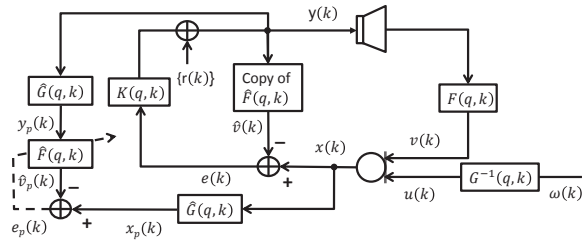


Fig. 2: AFC-PEM model

2. AFC IN HEARING AIDS

A traditional AFC model is presented in Fig. 1. The microphone signal is an addition of the incoming signal $u(k)$ and the feedback signal $v(k)$, i.e.,

$$x(k) = u(k) + v(k), \tag{1}$$

where k is the discrete-time index. The feedback signal is defined as $v(k) = \mathbf{f}^T(k)\,\mathbf{y}(k)$, where $\mathbf{f}(k) = \left[f_0(k), f_1(k), \ldots, f_{L_f-1}(k)\right]^T$ is a FIR filter of length L_f and $\mathbf{y}(k)$ is a vector of loudspeaker signal, $\mathbf{y}(k) = [y(k), y(k-1), \ldots, y(k-L_f+1)]^T$. L_f is the length of the feedback channel $F(q,k)$. The filter $F(q,k)$ can be represented as a polynomial transfer function in q, i.e., $F(q,k) = \mathbf{f}^T(k)\,\mathbf{q}$ with $\mathbf{q} = [1, q^{-1}, \ldots, q^{-L_f+1}]$. The error signal is yielded by subtracting the estimate of feedback signal from the microphone signal, i.e.,

$$e(k) = x(k) - \hat{\mathbf{f}}^T(k)\,\mathbf{y}(k), \tag{2}$$

where $\hat{\mathbf{f}}(k)$ is an estimate of $\mathbf{f}(k)$. The loudspeaker is formed by amplifying and delaying the error signal in the forward path,

$$y(k) = K(q,k)\,e(k) + r(k). \tag{3}$$

We assume that the forward path consists of an amplifier with a gain of $|K|$ and a delay $d_k \geqslant 1$ sample, i.e., $K(q,k) = |K|\,q^{-d_k}$. A white Gaussian probe noise $r(k)$ might be coupled into the loudspeaker signal before adaptive process to further decouple the loudspeaker and microphone signals.

3. PROPOSED PEM-APA

In this section, we describes details on how to apply a classical APA to the AFC-PEM to improve the convergence rate while remaining low steady-state error. We use the same model of the AFC-PEM for both adaptive filtering algorithms, NLMS and APA. Fig. 2 represents an acoustic feedback cancellation model using prediction error method (AFC-PEM) which has been introduced in [2, 4, 15]. In this method, the incoming signal is assumed to be modeled by an auto-regressive (AR) process, i.e., it can be computed as

$$u(k) = G^{-1}(q,k)\,w(k), \tag{4}$$

where $G^{-1}(q,k)$ is an all-pole filter and $w(k)$ is a white Gaussian noise sequence. Normally, the inverse of that filter is an FIR filter whose coefficients are estimated by Levinson-Durbin algorithm [16] or Burg-Lattice algorithm [2, 17]. Then the estimated filter $\hat{G}(q,k)$ is used to pre-whiten the loudspeaker and microphone signals, i.e.,

$$x_p(k) = \hat{G}(q,k)\,x(k) \tag{5}$$

$$y_p(k) = \hat{G}(q,k)\,y(k). \tag{6}$$

Commonly, the well-known NLMS algorithm is found to be an attractive choice for the AFC-PEM due to its simplicity and efficiency, but the convergence rate of the system deteriorates when the incoming signals are spectrally coloured.

The adaptive filter coefficients of the feedback canceller $\hat{F}(q,k)$ in the PEM-NLMS can be estimated as follows,

$$\hat{\mathbf{f}}(k) = \hat{\mathbf{f}}(k-1) + \frac{\mu}{(||\mathbf{y}_p(k)||_2^2 + \delta)}\mathbf{y}_p(k)\,e_p(k), \tag{7}$$

where $\mathbf{y}_p(k) = \left[y_p(k), \ldots, y_p\left(k - L_{\hat{f}} + 1\right)\right]^T$ represents the vector of the pre-whitened loudspeaker signal and $L_{\hat{f}}$ is the length of the estimated feedback path.

We propose to employ an affine projection algorithm which provides a superior convergence rate compared to NLMS algorithm, especially for speech incoming signal in the AFC-PEM. In the proposed method, PEM-APA, P most recent input vectors are used to estimate the coefficients of an adaptive filter, where P is a projection order. Thus, the input vectors of the feedback canceller $\hat{F}(q,k)$ in the PEM-APA can be defined as follows

$$\mathbf{Y}_p(k) = [\mathbf{y}_p(k), \mathbf{y}_p(k-1), \ldots, \mathbf{y}_p(k-P+1)]. \tag{8}$$

The vector $\mathbf{e}_p(k) = [e_p(k), \ldots, e_p(k-P+1)]^T$ denotes a pre-whitened error signal which is computed by subtracting the pre-whitened microphone signal from the estimated feedback signal, i.e.,

$$\mathbf{e}_p(k) = \mathbf{x}_p(k) - \mathbf{Y}_p^T(k)\,\hat{\mathbf{f}}(k), \tag{9}$$

where $\mathbf{x}_p(k) = [x_p(k), x_p(k-1), \ldots, x_p(k-P+1)]^T$ is a vector of the pre-whitened microphone signal. In the PEM-APA the adaptive filter coefficients of the feedback canceller $\hat{F}(q,k)$ can be estimated as follows,

$$\hat{\mathbf{f}}(k) = \hat{\mathbf{f}}(k-1) + \mu \mathbf{Y}_p(k) \left[\mathbf{Y}_p^T(k)\mathbf{Y}_p(k) + \mathbf{I}_P\delta\right]^{-1} \mathbf{e}_p(k), \tag{10}$$

where $\mathbf{I}_P$ is an $(P \times P)$ identity matrix, δ is a small positive value to avoid division by zero and μ is a fixed step size used to control the update of $\hat{\mathbf{f}}(k)$. By selecting a small projection order P, e.g., $P = 2$, the computational complexity of the PEM-APA increases only $\left(5L_{\hat{f}} + 8\right)$ multiplications and $\left(5L_{\hat{f}} + 9\right)$ additions per iteration compared to the PEM-NLMS [17].

4. SIMULATION RESULTS

In this section, the proposed PEM-APA is evaluated for a speech incoming signal. A dummy head with adjustable ear canals [18] in conjunction with a two-microphone behind-the-ear hearing aid and open-fitting ear molds [19] was used to measure acoustic feedback paths in two scenarios free-field $(F_1(f))$ and with a telephone receiver $(F_2(f))$ placed close to the hearing aid. The sampling frequency was $f_s = 16$kHz and the length of the measured feedback paths was $L_f = 100$. Fig. 3 shows the amplitude characteristic of the impulse responses (IR) used in simulations. A speech incoming signal created by concatenating male and female speech used in [20] was selected as the incoming signal $u(k)$. All simulation were implemented with 50s long speech incoming signal and with a sudden change from the free-field to the telephone-near feedback path at the 25th second. Misalignment and added stable gain were used in performance evaluation of the AFC methods. The normalized misalignment (MIS) is denoted as

$$MIS = 10\log_{10}\left(\frac{||\mathbf{f} - \hat{\mathbf{f}}||_2^2}{||\mathbf{f}||_2^2}\right), \tag{11}$$

while the added stable gain (ASG) is defined as [4, 21]

$$\begin{aligned} \text{ASG} &= 10\log_{10}\frac{1}{\max_\Omega |F(\Omega) - \hat{F}(\Omega)|^2} \\ &- 10\log_{10}\frac{1}{\max_\Omega |F(\Omega)|^2}, \end{aligned} \tag{12}$$

where $F(\Omega)$ and $\hat{F}(\Omega)$ are the frequency responses of measured and estimated acoustic feedback paths at the normalized frequency, respectively. We used the following settings for all simulations. The forward path gain and delay of the hearing aid were set to $|K| = 30\,dB$ and $d_k = 96$ samples, respectively. The feedback canceller path was delayed by 1 sample. Also, $L_{\hat{f}} = 64$, $\mu = 0.001$ were chosen for the time-domain implementation. The order L_p of the prediction-error filter $G(q,k)$ was 17 and was updated using the Levinson-Durbin algorithm every 10 ms in both the PEM-NLMS and the PEM-APA. For the PEM-APA, the smallest projection order of 2 was selected. Fig. 4 shows the simulation results for the AFC-PEM using NLMS and APA for the case of no probe noise. The convergence performance of the PEM-APA outperforms that of the PEM-NLMS. When the feedback path changes from free-field feedback path to telephone-near feedback path, the PEM-APA can track that change much quicker than the PEM-NLMS. The averages of misalignment $(\overline{MIS})$ and added stable gain $(\overline{ASG})$ are evaluated in Table 1. It can be seen that the PEM-APA yields significant improvements in both $\overline{ASG}$ and $\overline{MIS}$ compared to the PEM-NLMS.

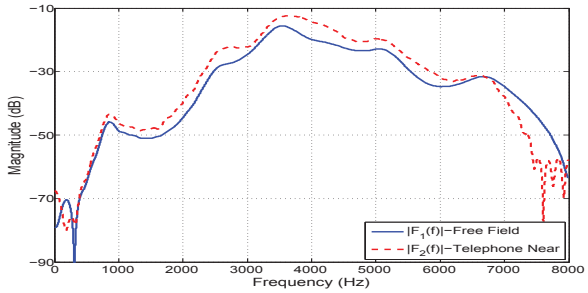
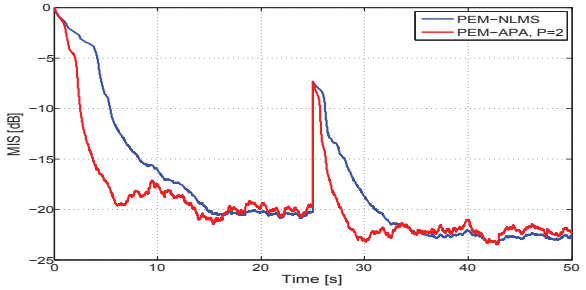


Fig. 3: Characteristics of measured feedback paths

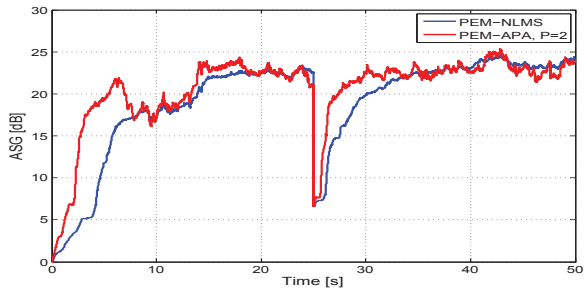
Fig. 5-6 illustrate the performances of the PEM-NLMS and the PEM-APA in the case of using a probe noise (PN). In this case, a white Gaussian noise with SNR = 25 dB or SNR = 20 dB is injected into the loudspeaker signal. All other parameters are set the same as the case of no probe noise. The probe noise provides further decorrelation between loudspeaker and incoming signals, thus the better performance can be achieved for both mentioned systems. However, the PEM-APA-PN still outperforms the PEM-NLMS-PN in both terms of misalignment and added stable gain. It converges much quicker and tends to have lower steady-state error. Furthermore, the PEM-APA-PN provides more added stable gain, up to 4 dB for free-field feedback path and up to 2 dB for telephone-near feedback path for the case SNR = 25 dB, and up to 5 dB gain for both mentioned feedback paths for the case SNR = 20 dB compared to the PEM-NLMS-PN. It also obtains very high tracking rate when the feedback path changes. The advantages of the PEM-APA-PN are confirmed again in the evaluation of $\overline{MIS}$ and $\overline{ASG}$ as in Table 2-Table 3.

Table 1: PEM-NLMS and PEM-NLMS-APA with $P = 2$, $\mu = 0.001$, $L_p = 17$, speech incoming signal.

AFC methods	$\overline{ASG}$	$\overline{MIS}$
PEM-NLMS	19.0355	-17.8401
PEM-APA	**20.5528**	**−19.3757**

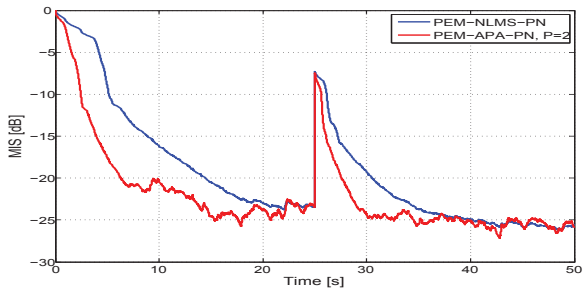


(a) Misalignment

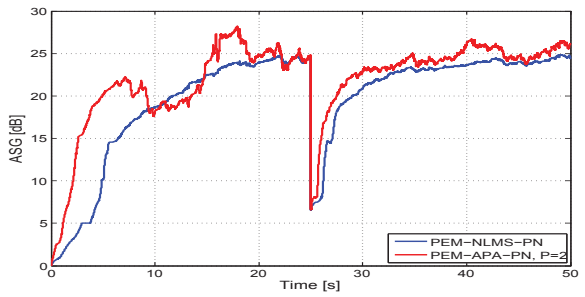


(b) Added Stable Gain

Fig. 4: Performance of PEM-NLMS, PEM-APA (P=2) without a probe noise $\mu = 0.001$, $L_p = 17$, speech incoming signal.

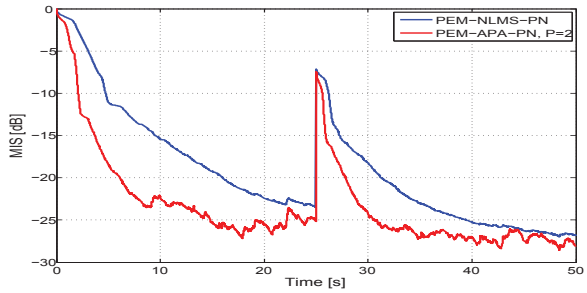


(a) Misalignment

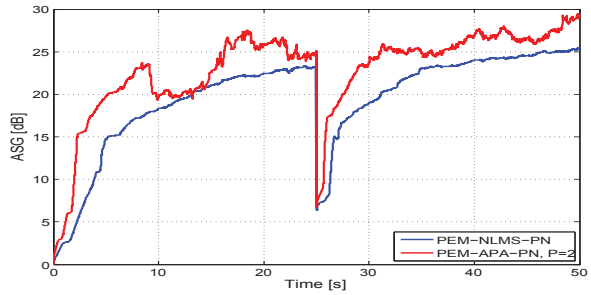


(b) Added Stable Gain

Fig. 5: Performance of PEM-NLMS, PEM-APA ($P = 2$) with a probe noise $SNR = 25dB$, $\mu = 0.001$, $L_p = 17$, speech incoming signal.



(a) Misalignment



(b) Added Stable Gain

Fig. 6: Performance of PEM-NLMS, PEM-APA ($P = 2$) with a probe noise $SNR = 20dB$, $\mu = 0.001$, $L_p = 17$, speech incoming signal.

Table 2: PEM-NLMS-PN, PEM-APA-PN with $P = 2$, $\mu = 0.001$, $L_p = 17$, $SNR = 25dB$, speech incoming signal.

AFC methods	$\overline{ASG}$	$\overline{MIS}$
PEM-NLMS-PN	19.7654	-19.1521
PEM-APA-PN	**21.9399**	$\mathbf{-21.8836}$

Table 3: PEM-NLMS-PN, PEM-APA-PN with $P = 2$, $\mu = 0.001$, $L_p = 17$, $SNR = 20dB$, speech incoming signal.

AFC methods	$\overline{ASG}$	$\overline{MIS}$
PEM-NLMS-PN	19.5243	-18.9809
PEM-APA-PN	**22.8104**	$\mathbf{-23.1720}$

5. CONCLUSION

This paper proposes a simple but efficient manner to implement the APA in the AFC-PEM. The PEM is known as a promising method to decorrelate the input signals, whereas the APA is a powerful algorithm to improve the convergence rate. Thus, the proposed method shows much faster convergence rate, much quicker tracking rate, lower $\overline{MIS}$ and higher $\overline{ASG}$ than the PEM-NLMS. Moreover, the PEM-APA can work well for hearing aids with/without a probe noise.

6. REFERENCES

[1] J. Hellgren, "Analysis of feedback cancellation in hearing aids with filtered-x lms and the direct method of closed loop identification," *IEEE Transactions on Speech and Audio Processing*, vol. 10, no. 2, pp. 119–131, 2002.

[2] A. Spriet, I. Proudler, M. Moonen, and J. Wouters, "Adaptive feedback cancellation in hearing aids with linear prediction of the desired signal," *IEEE Transactions on Signal Processing*, vol. 53, no. 10, pp. 3749–3763, 2005.

[3] G. Rombouts, T. Van Waterschoot, and M. Moonen, "Robust and efficient implementation of the pem afrow algorithm for acousic feedback cancellation," *Journal of the Audio Engineering Society*, vol. 55, no. 11, pp. 955–966, 2007.

[4] A. Spriet, S. Doclo, M. Moonen, and J. Wouters, "Feedback control in hearing aids," in *Springer Handbook of Speech Processing*. Springer, 2008, pp. 979–1000.

[5] K. Ozeki and T. Umeda, "An adaptive filtering algorithm using an orthogonal projection to an affine subspace and its properties," *Electronics and Communications in Japan (Part I: Communications)*, vol. 67, no. 5, pp. 19–27, 1984.

[6] M. Tanaka, Y. Kaneda, S. Makino, and J. Kojima, "A fast projection algorithm for adaptive filtering," *IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences*, vol. 78, no. 10, pp. 1355–1361, 1995.

[7] F. Albu and C. Kotropoulos, "Modified gauss-seidel affine projection algorithm for acoustic echo cancellation." in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2005, pp. 121–124.

[8] J. Benesty, P. Duhamel, and Y. Grenier, "A multichannel affine projection algorithm with applications to multichannel acoustic echo cancellation," *IEEE Signal Processing Letters*, vol. 3, no. 2, pp. 35–37, 1996.

[9] Q. Liu, B. Champagne, and K. Ho, "On the use of a modified fast affine projection algorithm in subbands for acoustic echo cancelation," in *IEEE Digital Signal Processing Workshop Proceedings*, 1996, pp. 354–357.

[10] C. Paleologu, J. Benesty, and S. Ciochina, "A variable step-size affine projection algorithm designed for acoustic echo cancellation," *IEEE Transactions on Audio, Speech, and Language Processing*, vol. 16, no. 8, pp. 1466–1478, 2008.

[11] F. Albu, C. Paleologu, and S. Ciochina, "New variable step size affine projection algorithms," in *9th IEEE International Conference on Communications (COMM)*, 2012, pp. 63–66.

[12] K. Lee, Y.-h. Baik, Y. Park, D. Kim, and J. Sohn, "Robust adaptive feedback canceller based on modified pseudo affine projection algorithm," in *2011 Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, 2011, pp. 3760–3763.

[13] S. Lee, I.-Y. Kim, and Y.-C. Park, "Approximated affine projection algorithm for feedback cancellation in hearing aids," *Computer methods and programs in biomedicine*, vol. 87, no. 3, pp. 254–261, 2007.

[14] L. T. T. Tran, S. Nordholm, H. Dam, W. Yan, and C. Nakagawa, "Acoustic feedback cancellation in hearing aids using two microphones employing variable step size affine projection algorithms," in *2015 IEEE International Conference on Digital Signal Processing (DSP)*, 2015, pp. 1191–1195.

[15] A. Spriet, G. Rombouts, M. Moonen, and J. Wouters, "Adaptive feedback cancellation in hearing aids," *Journal of the Franklin Institute*, vol. 343, no. 6, pp. 545–573, 2006.

[16] J. R. Deller Jr, J. G. Proakis, and J. H. Hansen, *Discrete time processing of speech signals*. Prentice Hall PTR, 1993.

[17] A. H. Sayed, *Fundamentals of adaptive filtering*. John Wiley & Sons, 2003.

[18] M. Hiipakka, M. Tikander, and M. Karjalainen, "Modeling the external ear acoustics for insert headphone usage," *Journal of the Audio Engineering Society*, vol. 58, no. 4, pp. 269–281, 2010.

[19] T. Sankowsky-Rothe, M. Blau, H. Schepker, and S. Doclo, "Reciprocal measurement of acoustic feedback paths in hearing aids," *The Journal of the Acoustical Society of America*, vol. 138, no. 4, pp. EL399–EL404, 2015.

[20] C. R. C. Nakagawa, S. Nordholm, and W.-Y. Yan, "New insights into optimal acoustic feedback cancellation," *IEEE Signal Processing Letters*, vol. 20, no. 9, pp. 869–872, 2013.

[21] J. Hellgren and U. Forssell, "Bias of feedback cancellation algorithms in hearing aids based on direct closed loop identification," *IEEE Transactions on Speech and Audio Processing*, vol. 9, no. 8, pp. 906–913, 2001.

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